

# Econ 30065: Microeconomics

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L6

# Utility

- Utility is a numeric value indicating the consumer's **relative** well being from consuming a bundle of goods.
- Higher utility indicates greater satisfaction than lower utility.
- A **utility function** is a mathematical formula that assigns a utility value to each consumption bundle:  
 $U(X,Y)=f(X,Y)$
- Example:  $U(F,C)=2F+5(F \times C)$   
A basket of Fries=2 and Chicken=1 gives a utility:  
$$U(2,1)=2 \times 2 + 5(2 \times 1) = 14$$

# Ordinal vs. Cardinal Utility

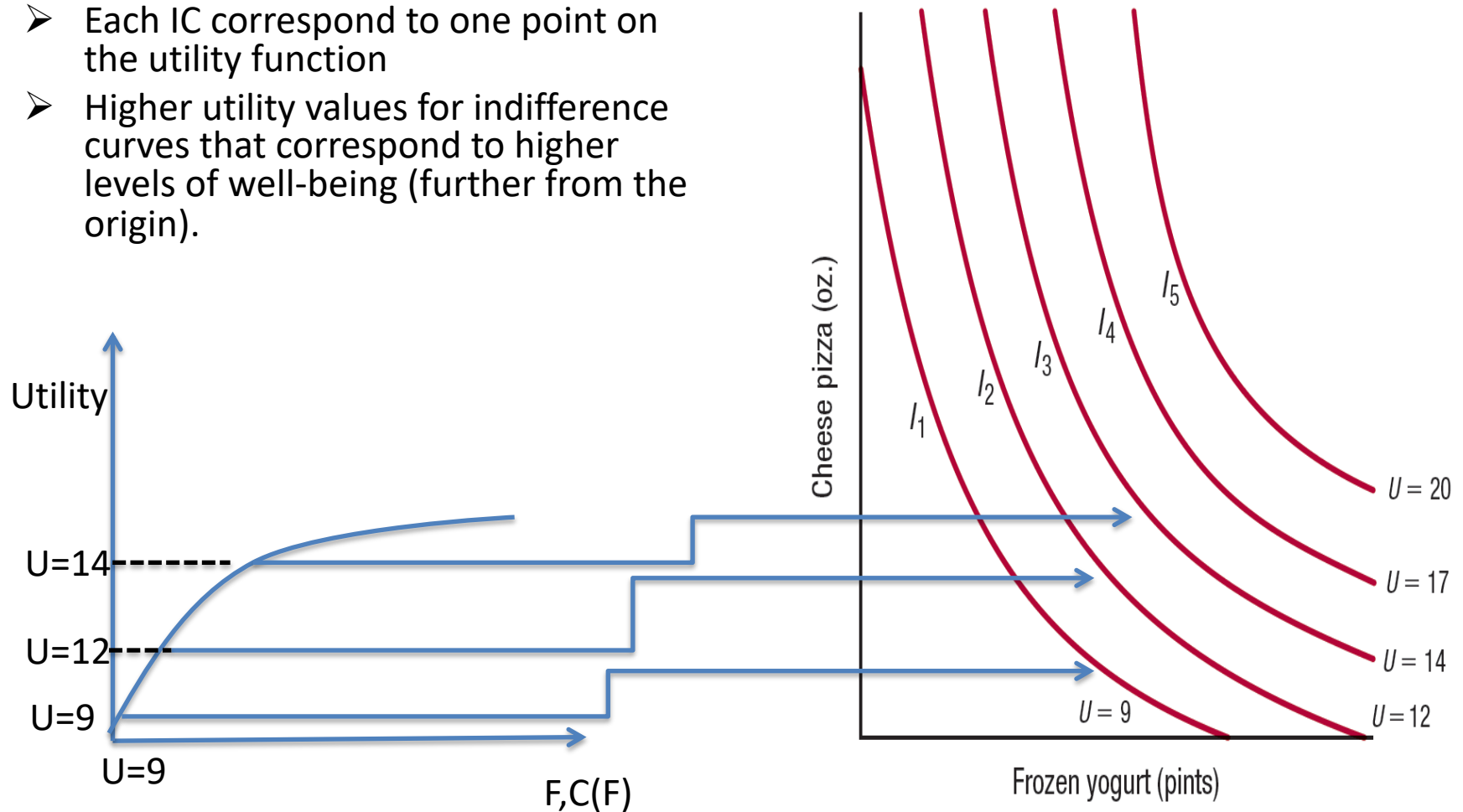
- **Ordinal information** allows us to determine only whether one alternative is better or worse than another.
- **Cardinal information** tells us something about the *intensity* of those preferences.
- In modern microeconomic theory, utility functions are only intended to summarize ordinal information, rejecting old cardinal interpretations of utility.
  - Example: if one consumption bundle has a utility value of 10 and a second has a utility value of 5, we know the consumer prefers the first to the second, but it does not necessarily make him twice as happy.

# From Preferences to Utility

- Utility functions: starting from preferences to summarize consumers' behavior.
- Start from preferences: give same utility to bundles on same indifference curves
- Higher utility to bundles on IC further from the origin.
- You can go from IC to utility and viceversa

# From Indifference Curves to Utility Function

- Each IC correspond to one point on the utility function
- Higher utility values for indifference curves that correspond to higher levels of well-being (further from the origin).



# Example

- Robert enjoys reading books and watching movies. His utility function is  $U(M,B)=2M \times B$ , where  $M$  is the number of movies and  $B$  is the number of books enjoyed during a month. What do these curves look like?
- Fix  $U$
- $B = \underline{U} / 2M$
- Convex curves

# Example

- Marta enjoys reading books and watching movies. Her utility function is  $U(M,B)=M \times B^2$ , where  $M$  is the number of movies and  $B$  is the number of books enjoyed during a month. How does Marta rank the following bundles?
- 1) 4 movies and 5 books
- 2) 10 movies and 4 books
- 3) 25 movies and 2 books
- 4) 40 movies and 1 books

# Solution

- Find Utilities:
- Bundle 1) gives  $U=4 \times 5^2=100$
- 2)  $U=10 \times 4^2=160$
- 3)  $U=25 \times 2^2=100$
- 4)  $U=40 \times 1=40$
- Thus the second bundle is the preferred one



# Utility Function and MRS

MRS can be rewritten as:

$$MRS_{XY} = \frac{MU_X}{MU_Y}$$

Where  $MU_X$ , the marginal utility of good X, is the change in the consumer's utility resulting from the addition of a very small amount of that good:

$$MU_X = \frac{\Delta U_X}{\Delta X}$$

Same for  $MU_Y$ .

# Example continued

- What is Robert marginal utility from movies?  
And from books?
- *take derivative of utility wrt Movies:*
  - $MU_M = 2B$
- *take derivative of utility wrt Books:*
  - $MU_B = 2M$
- What is the MRS for movies with books?
  - $MU_m / MU_B = B / M$

# Some Common Utility Functions

- Cobb-Douglas utility functions

$$U(X,Y)=X^aY^b$$

# Cobb-Douglas Utility Functions

- Two families of indifference curves for Cobb-Douglas utility functions
- Declining MRS
- $MRS = (a/b)(Y/X)$
- Slope depends on  $a$  and  $b$

